

Report
Of
Java Full Stack Developer Virtual Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410031588
Name of Student: ROHAN SINGH

Batch: 2026-27

CANDIDATE'S DECLARATION

I, Rohan Singh, Roll No. 2410031588, do hereby declare that the internship report titled “Java Full Stack Developer Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Rohan Singh

Roll No.: 2410031588

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Java Full Stack Developer Virtual Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE, the National Internship Portal and EduSkills Academy for organizing and supporting this virtual internship, and my mentors and faculty at the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous guidance and encouragement throughout the duration of this internship.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Rohan Singh
Roll No.: 2410031588

INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Rohan Singh

IILM University, Greater Noida

has successfully completed the 8-weeks

Java Full Stack Developer Virtual Internship

During August - October 2026

Supported by  **EduSkills
ACADEMY**



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 4c05a4ae6a641a9f2cf0
Student ID: STU6a858aa9bcd61787136681



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during the 8-week “Java Full Stack Developer Virtual Internship”, organized under the National Internship Portal by the All India Council for Technical Education (AICTE), Ministry of Education, Government of India, in association with EduSkills Academy. The internship was completed from August to October 2026 as part of the curriculum requirement for the degree of Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida. The program was designed to provide hands-on, industry-oriented exposure to full stack Java development, covering both the fundamentals of programming and the practical skills required to design, build and deploy modern web applications.

Full stack development refers to the ability to work across every layer of a software application — from the user interface and client-side logic, through the server-side business logic, to the underlying database. Java, being a mature, platform-independent and widely adopted language in enterprise software development, forms the backbone of this internship’s curriculum, supported by frameworks and tools that are standard in the industry today.

4.2 Organization Profile

The internship was facilitated by EduSkills Academy, a skilling and training organization that partners with AICTE and various industry leaders to deliver virtual internships and certification programs to engineering students across India under the banner “Nation Building Through Skills.” EduSkills Academy works in collaboration with the Ministry of Education’s National Internship Portal to bridge the gap between academic learning and industry requirements by offering structured, mentor-guided virtual internships in emerging technology domains such as full stack development, cloud computing, artificial intelligence and cybersecurity.

As an AICTE-recognized initiative, the program follows a standardized curriculum and evaluation framework, ensuring that participating students across different institutions receive a consistent, quality learning experience along with an industry-recognized certificate of completion upon successful evaluation.

Learners were supported through self-paced online modules, recorded lectures and reference material hosted on the EduSkills platform, with periodic quizzes and a final assessment used to grade performance. This mentor-guided, self-paced format allowed

students to balance the internship alongside their regular academic coursework while still receiving structured milestones and timely feedback on their progress.

4.3 Problem Statement

Traditional academic curricula often emphasize theoretical concepts of programming without providing sufficient exposure to the complete workflow of building a real-world web application. Students frequently learn front-end and back-end concepts in isolation, without understanding how these layers integrate with a database to form a fully functional system. There was, therefore, a need for a structured, practical learning path that would take a student through the entire development lifecycle of a Java-based web application — from designing the user interface, to implementing business logic on the server, to persisting and retrieving data — while following industry-standard practices.

4.4 Project Objectives

- To understand and apply core Object-Oriented Programming principles using Java.
- To design responsive front-ends using HTML, CSS and JavaScript.
- To build REST APIs and server-side logic using Spring and Spring Boot, integrated with MySQL via JDBC/Hibernate.
- To gain hands-on experience with professional tools such as Git, Maven and Postman.
- To develop and test a complete full stack application integrating all layers.

4.5 Scope of the Project

The scope of the internship encompassed the complete spectrum of full stack Java development: front-end technologies for the presentation layer, core and advanced Java for application logic, the Spring/Spring Boot framework for scalable back-end services and REST APIs, and relational database management using MySQL. The modules progressed from foundational concepts to advanced topics such as API design, security basics and deployment, giving a holistic view of how a modern web application is built and delivered.

4.6 Technologies and Tools Used

- Programming Language: Java (Core Java and Object-Oriented Programming concepts)
- Front-End: HTML5, CSS3, JavaScript

- Back-End Framework: Spring, Spring Boot
- Data Access: JDBC, Hibernate/JPA
- Database: MySQL
- Build and Dependency Management: Maven
- Version Control: Git and GitHub
- Development Environment: Eclipse / IntelliJ IDEA / Visual Studio Code
- API Testing: Postman

4.7 System Architecture

The applications developed during the internship followed a layered (n-tier) architecture typical of enterprise Java applications. The Presentation Layer, built with HTML, CSS and JavaScript, handles user interaction and displays data received from the server. The Business/Service Layer, implemented using Spring and Spring Boot, contains the core application logic and exposes functionality through RESTful web service endpoints. The Data Access Layer, built using JDBC and Hibernate/JPA, manages communication between the application and the underlying MySQL database. This separation of concerns ensures that each layer can be developed, tested and maintained independently, which is a key principle of scalable software design.

4.8 Methodology

The internship followed a structured, module-based methodology delivered over eight weeks. Each week focused on a specific stage of full stack development, beginning with an introduction to Java fundamentals and Object-Oriented Programming, followed by front-end web development, then back-end development with Spring Boot, database integration, and finally the design and testing of RESTful APIs. Learning was reinforced through self-paced video lectures, reading material, hands-on coding exercises and periodic assessments, with concepts applied progressively to build towards a complete working application by the end of the program.

4.9 Expected Outcomes

- A solid conceptual and practical understanding of Java and Object-Oriented Programming.
- The ability to independently design and build responsive web front-ends.
- Competence in developing REST APIs and back-end services using Spring Boot.
- Practical experience in connecting a Java application to a relational database.

- Improved problem-solving, debugging and independent-learning skills.
- Readiness to take up entry-level roles or further projects in full stack Java development.

4.10 Certificates of Completion and Other Communication/Mail Proof

The Certificate of Virtual Internship issued jointly by the Ministry of Education, AICTE, the National Internship Portal and EduSkills Academy on successful completion of the 8-week “Java Full Stack Developer Virtual Internship” (August–October 2026) is attached in Chapter 3 of this report (Certificate ID: 4c05a4ae6a641a9f2cf0, Student ID: STU6a858aa9bcbd61787136681). The certificate records a final grade of “O – Outstanding.”

CHAPTER 5

BIBLIOGRAPHY / REFERENCES

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